

Balancing Robot

14/05/2026

Balancing Robot				14/05/2026							21/05/2026							28/05/2026							04/06/2026							11/06/2026							
				14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
TASK	START	DAYS	END	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	
Line Following																																							
Initial Sensor Research	18/05/2026	8	25/05/2026																																				
Circuit to read from one IR sensor	27/05/2026	3	29/05/2026																																				
Prototype IR array on breadboard	28/05/2026	5	01/06/2026																																				
Solder circuit to stripboard	01/06/2026	3	03/06/2026																																				
IR mount design 1	04/06/2026	2	05/06/2026																																				
Create line track with black tape	01/06/2026	1	01/06/2026																																				
Line following test on stabilised robot	05/06/2026	4	08/06/2026																																				
IR mount design 2	08/06/2026	2	09/06/2026																																				
Line following test on balancing robot	10/06/2026	3	12/06/2026																																				
Head Unit Construction																																							
Pi camera mount design 1	09/06/2026	2	10/06/2026																																				
Pi camera mount design 2	10/06/2026	2	11/06/2026																																				
Monitoring of Power and Battery Status																																							
Design and simulate circuits in LTSpice	08/06/2026	2	09/06/2026																																				
Prototype circuits on breadboard	10/06/2026	1	10/06/2026																																				
Testing and adjusting with PSU	10/06/2026	2	11/06/2026																																				
Code to read output with ADC	12/06/2026	1	12/06/2026																																				
Soldering circuits to stripboards	12/06/2026	1	12/06/2026																																				
Static Balancing and Dynamic Stability																																							
Assemble robot and test starter code	14/05/2026	2	15/05/2026																																				
Tuning PID inner loop through webpage	18/05/2026	5	22/05/2026																																				
Accelerometer and Gyro calibration	20/05/2026	1	20/05/2026																																				
PD Yaw controller for spot turns	26/05/2026	3	28/05/2026																																				
Tune velocity outer loop	26/05/2026	13	07/06/2026																																				
Implement turning whilst driving	08/06/2026	2	09/06/2026																																				
Raspberry Pi																																							
Setup RPi with Picam	14/05/2026	2	15/05/2026																																				
Webpage to view video feed	18/05/2026	4	21/05/2026																																				
Interface RPi with ESP32 via UART	06/06/2026	3	08/06/2026																																				
Colour detection library	09/06/2026	4	12/06/2026																																				
TOF Sensors (obstacle/maze)																																							
Researching and selecting sensors	19/05/2026	4	22/05/2026																																				